Cube-Sum Permutations: Exact Membership, Multiplicity, and an Exponential Lower Bound

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ such that $i+p_i$ is a perfect cube for every index i. We give a computer-assisted resolution of the existence conjecture recorded as OEIS A006063, as well as an exact membership classification for its related sequence A096680: $P(n) > 0$ for every $n \geq 173$, whereas $P(173) = P(174) = 1$ and $P(n) \geq 3$ for every $n \geq 175$. Exact Hall-set certificates determine all earlier existence failures. An interval-reflection lemma reduces the unbounded existence theorem to only 109 explicit core permutations; alternating-cycle certificates establish multiplicity. Finally we prove the nontrivial quantitative bound

$$P(n) \geq 3^{\max\{0,\lfloor(\lfloor(2n)^{1/3}\rfloor-27)/6\rfloor\}} = \exp\bigl(\Omega(n^{1/3})\bigr).$$

All finite certificates, exact checkers, and an independent matching implementation accompany this note. Priority relative to the wider literature and independent certificate review remain open.

1 The reflection principle

Write $\mathcal{C} = \{k^3 : k = 2, 3, \dots\}$ and let $P(n)$ be the number of cube-sum permutations of $\{1,\dots,n\}$. By convention the empty permutation is permitted only in the general extension lemma; our classification concerns $n \geq 1$. Let $\mathcal{G} = \{n \geq 1 : P(n) > 0\}$.

Lemma 1 (Reflection and counting)**.** *If $0 \leq m < n$ and $c = m + n + 1$ is a cube, every valid permutation p on $\{1,\dots,m\}$ extends to a valid permutation q on $\{1,\dots,n\}$ via*

$$q_i = \begin{cases} p_i, & i \leq m, \\ c - i, & m < i \leq n. \end{cases}$$

For different admissible cubes c, the images of these extensions are disjoint. In particular,

$$P(n) \geq \sum_{\substack{c \in \mathcal{C} \\ n+2 \leq c \leq 2n}} P(c-n-1), \tag{1}$$

with the convention $P(0) = 1$.

Proof. Because $c = m+n+1$, reflection $i \mapsto c-i$ permutes precisely the new interval $\{m+1,\dots,n\}$, and $i+q_i = c$ there. Old indices retain their cube sums. The embedding is injective for each c and two different c yield different last entries $q_n = c - n$. The claimed sum counts disjoint families. □

Lemma 2 (The cube-gap bound)**.** *For each $n \geq 365$ some cube c satisfies $n + 174 \leq c \leq 2n$. For each $n \geq 556$ some cube c satisfies $n + 176 \leq c \leq 2n$.*

Proof. For $365 \leq n \leq 555$, take $c = 729$. For $n \geq 556$, let k be the least integer with $k^3 \geq n + 176$; then $k \geq 10$ and $n \geq (k-1)^3 - 175$. Thus

$$2n - k^3 \geq k^3 - 6k^2 + 6k - 352 \geq 0.$$

The final polynomial equals 108 at $k = 10$ and has strictly positive successive differences $3k^2 - 9k + 1$ for $k \geq 10$. Hence the second assertion holds and implies the first for $n \geq 556$. □

2 Finite certificates and exact membership

Our finite data are provided as UTF-8 JSON arrays of integers. A valid witness p is checked solely by

$$\mathrm{sort}(p) = (1, \ldots, n), \qquad \forall i \quad i + p_i \in \mathcal{C}.$$

For a negative instance, let $N_n(S) = \{j \in [1, n] : \exists i \in S,\ i + j \in \mathcal{C}\}$. Hall's marriage theorem gives the immediate obstruction

$$|N_n(S)| < |S| \quad \Longrightarrow \quad P(n) = 0. \tag{2}$$

The file `exact_classification_below_173.json` provides 48 explicit valid permutations and 124 explicit Hall-deficient subsets, exactly partitioning $1 \le n \le 172$. Every assertion is directly checked by `verify_extended.py`. For instance, for $n = 172$ the 27 left indices

$$S = \{1, 5, 8, 13, 20, 31, 35, 38, 42, 43, 45, 92, 96, 99,$$
$$104, 111, 122, 126, 129, 133, 134, 136, 153, 157, 160, 165, 172\}$$

have precisely 26 distinct possible right neighbors. Thus $172 \notin \mathcal{G}$, making the threshold below sharp.

For the unbounded tail, only the 109 witness sizes

$$173 \le n \le 255, \qquad 339 \le n \le 364 \tag{3}$$

require explicit stored permutations. Indeed for each $256 \le n \le 338$, put $m = 511 - n \in [173, 255]$; then $m + n + 1 = 512 = 8^3$, so Lemma 1 constructs the missing 83 cases. This compresses the previously saved 192 base cases by $83/192 > 43\%$. All 109 stored witnesses in `core_109.json`, and all 192 reconstructed witnesses, are checked with exact integer arithmetic. The earlier full 192-row file is retained as an additional independent-format audit source.

There is a further useful compression when the 48 verified positive sizes below 173 are reused. An ascending reflection-closure algorithm starts from those small solutions and, at each $173 \le n \le 364$, uses any already certified $m < n$ for which $m + n + 1$ is a cube. Precisely 131 lengths are obtained by reflection and only 61 further lengths need explicit witnesses. The 61 additional witnesses are archived in `core_61_with_small.json`, and `verify_extended.py` certifies this alternate compression independently of the 109-case presentation.

Theorem 1 (Complete A006063 membership)**.** *For every integer $n \ge 173$, $P(n) > 0$. Below 173, the complete membership list is*

$$\mathcal{G} \cap [1, 172] = \{7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,$$
$$103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121,$$
$$122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132,$$
$$133, 135, 147, 149, 150, 151, 152, 155, 156, 159, 171\}.$$

In sequence-index notation this gives

$$\boxed{\mathrm{A006063}(j) = j + 124 \quad (j \ge 49).}$$

Proof. The exact finite witnesses establish $173 \le n \le 364$, and the Hall certificates settle the smaller sizes. For $n \ge 365$, Lemma 2 gives a cube $c \in [n + 174, 2n]$. The predecessor $m = c - n - 1$ satisfies $173 \le m < n$. By strong induction $P(m) > 0$, and Lemma 1 yields $P(n) > 0$. There are precisely 48 positive indices below 173, so the 49th and all later sequence terms are consecutive. □

3 Multiplicity and the second OEIS classification

A cube-sum permutation p is a perfect matching in the bipartite graph with left and right vertices $1, \ldots, n$, where (i, j) is an edge exactly when $i + j \in \mathcal{C}$. For any matching p, form a directed graph D_p on the left vertices, putting $i \to j$ whenever $i \ne j$ and (i, p_j) is a cube-sum edge.

Lemma 3 (Alternating-cycle criterion)**.** *For a valid permutation p, there exists another distinct valid permutation if and only if D_p has a directed cycle.*

Proof. Along a directed cycle $i_1 \to \cdots \to i_t \to i_1$, replace p_{i_s} by $p_{i_{s+1}}$ (cyclic indices). Every moved pair is an allowed edge, and all right vertices remain distinct. Conversely, if $q \neq p$ is another valid permutation, the nontrivial cycles of the permutation $p^{-1} \circ q$ consist of directed cycles in D_p. Thus acyclicity is a rigorous uniqueness certificate. □

The file `second_solutions.json` records directed cycles, not complete duplicate permutations. Direct finite checking establishes: among the 48 positive lengths below 173, exactly

$$E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$$

have a directed alternating cycle; every other positive length below 173, as well as 173 and 174, has an acyclic D_p and therefore a unique permutation.

Furthermore, all 190 base cases $175 \leq n \leq 364$ have *two distinct* alternating-cycle certificates, archived as `second_solutions.json` and `third_solutions.json`. They yield three different permutations, counting the original matching. To extend the multiplicity assertion: for $365 \leq n \leq 553$, the cube 729 gives $m = 728 - n \in [175, 363]$. The two exceptional lengths $554, 555$ have separate pairs of alternating-cycle certificates obtained by extending the unique solutions at $174, 173$ respectively. For $n \geq 556$, the stronger part of Lemma 2 gives $m = c - n - 1 \geq 175$. Three distinct solutions for m remain distinct after reflection, finishing an induction.

Theorem 2 (Complete A096680 membership)**.** *The positive integers with at least two cube-sum permutations are exactly*

$$\boxed{E \ \cup \ \{175, 176, 177, \dots\}.}$$

Moreover $P(n) \geq 3$ for every $n \geq 175$, while $P(173) = P(174) = 1$. The sequence terms satisfy

$$\boxed{\text{A096680}(j) = j + 161 \quad (j \geq 14).}$$

Sharpness of the constant three. The bound $P(n) \geq 3$ for all $n \geq 175$ cannot be improved to four: $P(175) = 3$ exactly. For the stored matching at 175, the directed alternation graph has precisely one nontrivial strongly connected component,

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\},$$

and the other 165 vertices are singleton components. Every cycle of a permutation alternative lies in a strongly connected component, so the permanent reduces to the 10-by-10 component's allowed-edge matrix (including the matching self-edges). A direct bitmask dynamic program gives its permanent as 3. This computation is independently reproduced in `verify_extended.py` using exact integers.

4 Many permutations: an exponential cube-root bound

Besides qualitative multiplicity, the reflection family yields a strong quantitative statement. Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$, with the floor defined using integer comparisons (not floating point).

Lemma 4 (Three disjoint embeddings)**.** *If $K(n) = k \geq 27$, each of $k^3, (k-1)^3, (k-2)^3$ gives a reflection extension from a predecessor $m \geq 173$ with $m < n$, and all three predecessors obey $K(m) \geq k - 6$.*

Proof. By definition, $k^3 \leq 2n < (k+1)^3$. For the smallest of these cubes, set $m = (k-2)^3 - n - 1$. Then

$$2m > 2(k-2)^3 - (k+1)^3 - 2 = k^3 - 15k^2 + 21k - 19.$$

The right side exceeds $(k-6)^3$ for $k \geq 27$, since the difference is

$$3k^2 - 87k + 197 > 0 \quad (k \geq 27).$$

Thus $2m > (k-6)^3 \geq 21^3$, which implies $m \geq 173$ and $K(m) \geq k - 6$. Also $(k-2)^3 < k^3 \leq 2n$, so $m < n$. The other two cubes give larger predecessors and satisfy the same bounds. Distinct cubes yield disjoint matching families by Lemma 1. □

Theorem 3 (A096904 quantitative bound)**.** *For all integers $n \geq 173$, with $q(n) = \max\{0, \lfloor (K(n) - 27)/6 \rfloor\}$,*

$$\boxed{P(n) \geq 3^{q(n)}}.$$

In particular

$$\log P(n) \;\geq\; \frac{\log 3}{6}(2n)^{1/3} - O(1),$$

so $P(n)$ grows at least exponentially in $n^{1/3}$.

Proof. If $K(n) < 27$ then $q(n) = 0$, and Theorem 1 supplies $P(n) \geq 1$. For $K(n) \geq 27$, Lemma 4 provides three distinct extensions from smaller predecessors $m_i \geq 173$ with $K(m_i) \geq K(n) - 6$. Thus $q(m_i) \geq q(n) - 1$. By strong induction and the disjointness of the three extension images,

$$P(n) \geq \sum_{i=1}^{3} P(m_i) \geq 3 \cdot 3^{\max(0, q(n)-1)} \geq 3^{q(n)}.$$

The displayed logarithmic asymptotic follows immediately. □

5 An efficient constructive algorithm

The proof is also algorithmic. Given $n > 364$, choose the *least* cube $c = k^3 \geq n + 174$ and put $m = c - n - 1$. The cube-gap lemma ensures $173 \leq m < n$. Minimality gives

$$(k-1)^3 < n + 174 \leq k^3 \quad \Longrightarrow \quad 173 \leq m \leq 3k^2 - 3k + 173 = O(n^{2/3}).$$

Repeat until the remaining size is in the certified interval $173 \leq m \leq 364$, then fill each recorded interval by the explicit reflection $p_i = c - i$.

Proposition 1 (Linear-output construction)**.** *A valid cube-sum permutation of any $n \geq 173$ can be constructed in $O(n)$ unit-cost arithmetic operations, with $O(\log \log n)$ reflection layers.*

Proof. The predecessor bound $m = O(n^{2/3})$ implies $\log m \leq (2/3) \log n + O(1)$, so the number of iterations before the fixed-size base is $O(\log \log n)$. The returned intervals are disjoint and their union, together with the base interval, is $\{1, \ldots, n\}$. Therefore exactly n entries are written. Each exact integer cube root can be computed using $O(\log n)$ comparisons, which contributes $O(\log n \log \log n) = o(n)$ additional unit-cost operations. □

The accompanying `construct_linear.py` implements this procedure with exact integer cube roots, and can also display the recursion plan without allocating the full output.

6 Reproduction and limitations

All programs require only Python 3 standard-library modules:

```
python papers/A006063/verify.py
python papers/A006063/independent_match.py
python papers/A006063/verify_extended.py
python papers/A006063/construct_linear.py
```

The archived data consist of 109 core witnesses, the legacy 192-row all-base cross-check, 48 small positive permutations, 124 negative Hall sets, and compact pairs of alternating-cycle certificates. The infinite inductive and counting lemmas are mathematical arguments rather than extrapolations from numerical experiments.

The status is a *computer-assisted proof candidate* pending independent scrutiny of the finite certificate, its interpretation, and literature priority. The present manuscript does not compute the exact values of $P(n)$ in general, nor does it claim external peer review.

References

[1] N. J. A. Sloane et al., *A006063: A card-arranging problem*, `https://oeis.org/A006063`; existence conjecture attributed there to Robert Israel (2018).

[2] Ray Chandler, *A096680: Values with more than one cube-sum permutation*, `https://oeis.org/A096680`.

[3] Ray Chandler, *A096904: Number of cube-sum permutations*, `https://oeis.org/A096904`.

[4] P. Hall, On representatives of subsets, *J. London Math. Soc.* 10 (1935), 26–30.